Supplement: masking experiment for the imputation of burned area and duration

Table S1: Masking experiment: error of the imputed burned area (BA) and duration by fire-size class (defined by the observed value of the masked record). Bias = imputed − observed; MedAE = median absolute error; skill = $1 - \mathrm{RMSE}_{\ln}/\mathrm{RMSE}_{\ln}$ of the naive predictor (district–month median); coverage of the 5th–95th percentile interval of the 20 imputations (expected for exchangeable imputations: 81.4%).

Variable	Class (ha)	n	MedAE	Bias	$\mathrm{RMSE}_{\ln}$	Skill (%)	Coverage (%)
BA (ha)	All	3,940	0.19	−5.3	0.79	9.4	84.3
	< 1	2,926	0.09	+0.1	0.28	−48.0	88.9
	1–10	854	1.5	−1.3	0.94	5.9	79.2
	10–100	130	19.0	−22.6	2.67	9.1	28.5
	100–500	19	131.5	−52.3	4.02	15.7	26.3
	≥ 500	11	800.0	−1,451.2	4.59	30.5	27.3
Duration (min)	All	4,013	43.0	−93.0	1.11	−1.0	82.2
	< 1	3,003	36.5	−33.1	1.06	−2.9	83.7
	1–10	861	88.5	−172.7	1.15	−0.7	80.3
	10–100	121	194.5	−761.1	1.65	2.7	66.1
	100–500	18	576.3	−380.0	1.73	17.1	55.6
	≥ 500	10	3,237.2	−2,625.4	2.45	30.0	50.0

Table S2 gives the same statistics by data structure.

Table S2: Masking experiment by data structure (same statistics and units as Table S1).

Variable	Structure	n	MedAE	Bias	$\mathrm{RMSE}_{\mathrm{ln}}$	Skill (%)	Coverage (%)
BA (ha)	t1 (1980–1992)	566	0.50	−20.3	1.11	14.3	85.0
	t2 (1993)	77	0.14	+0.9	0.59	34.9	89.6
	t3 (1994–1995)	262	0.45	−7.0	0.87	3.3	89.3
	t4 (1996–2000)	678	0.18	−2.1	0.71	2.9	87.9
	t5 (2001–2025)	2,357	0.13	−2.5	0.71	7.9	82.4
Duration (min)	t1 (1980–1992)	568	44.3	−84.8	0.91	−1.3	89.6
	t2 (1993)	71	40.5	−24.6	0.72	−7.3	93.0
	t3 (1994–1995)	255	896.5	−767.1	2.22	1.3	38.8
	t4 (1996–2000)	719	37.0	−36.5	0.79	−4.3	92.1
	t5 (2001–2025)	2,400	41.0	−42.3	1.07	−1.4	81.8

Table S3 shows the observed durations by data structure, which help to explain the large duration errors in 1994–1995 (t3).

Table S3: Observed fire duration by data structure (records with a valid duration, excluding durations flagged as estimated or anomalous).

Structure	n	Median (min)	P90 (min)	Multiples of 60 min (%)	$\geq$ 12 h (%)
t1 (1980–1992)	112,907	95	300	19.6	2.6
t2 (1993)	15,928	75	200	11.5	0.9
t3 (1994–1995)	53,436	995.5	1,440	8.0	61.4
t4 (1996–2000)	143,793	80	230	6.6	1.1
t5 (2001–2025)	492,824	62	208	2.6	1.3